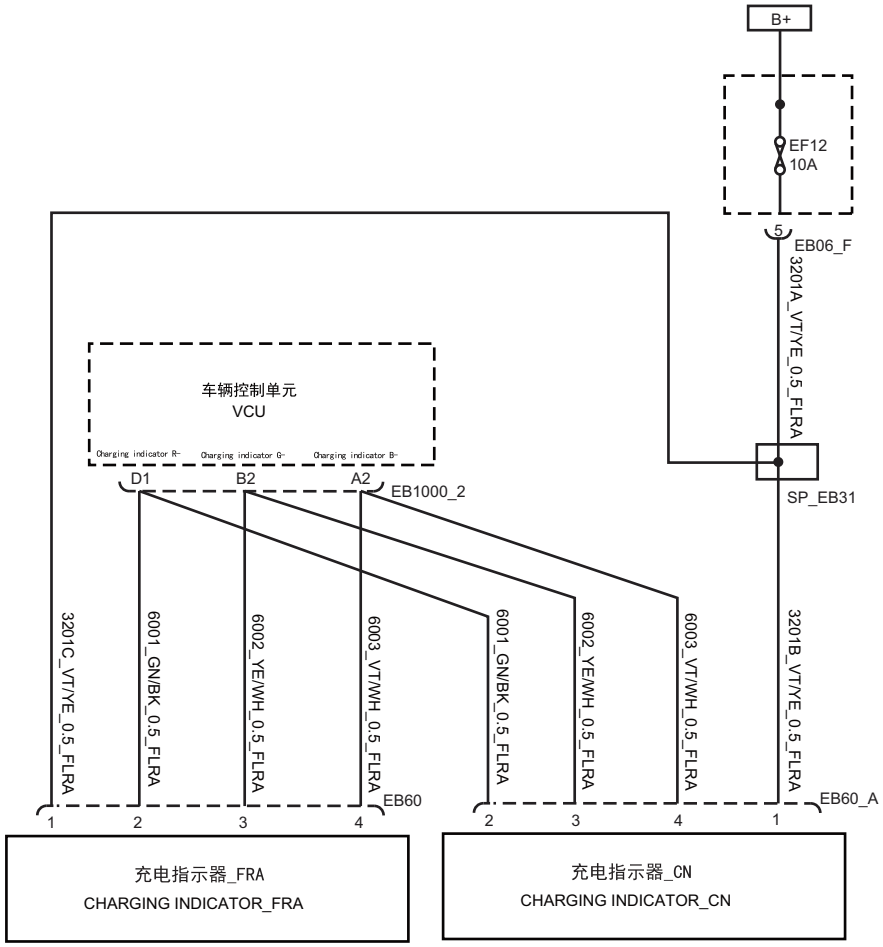
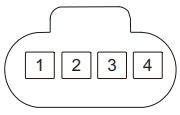
## P1CD7、 P1CD8、 P1CD9

| **Fault diagnosis code** |
| --- |
| P1CD7：DEM_stChrgLampGRlySBCfmd |
| P1CD8：DEM_stChrgLampGRlySGCfmd |
| P1CD9：DEM_stChrgLampGRlyOCCfmd |
| **Detection tools** |
| 1 Multimeter and diagnostic apparatus |
| **Possible causes** |
| • Circuit,terminal and adapting piece fault<br>• VCU module fault<br>• Charge indicator fault |
| **Conditions for fault code operation** |
| Turn the ignition switch to ON position. |
| **Conditions for fault code setting** |
| P1CD7 |
| Charge indicator G low level control relay control terminal short to battery |
| P1CD8 |
| Charge indicator G low level control relay control terminal short to ground |
| P1CD9 |
| Charge indicator G low level control relay control terminal open |
| **To set the effect of a fault code condition** |
| |
| **Description of circuit diagram** |
| The VCU module monitors charge indicator via low level reference circuit and signal circuit. |

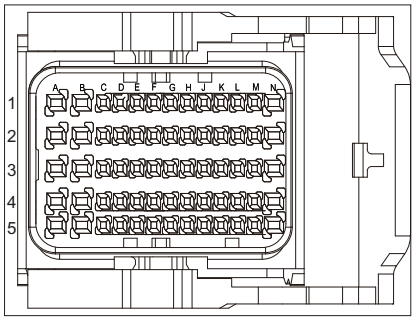
**Circuit diagram**



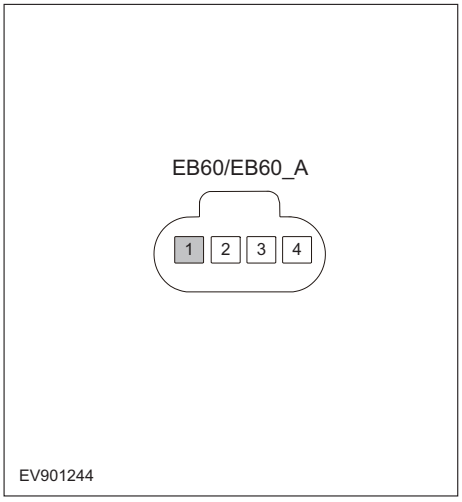

EB60/EB60_A

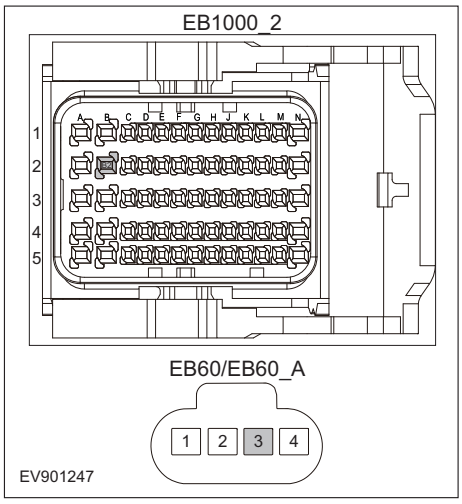


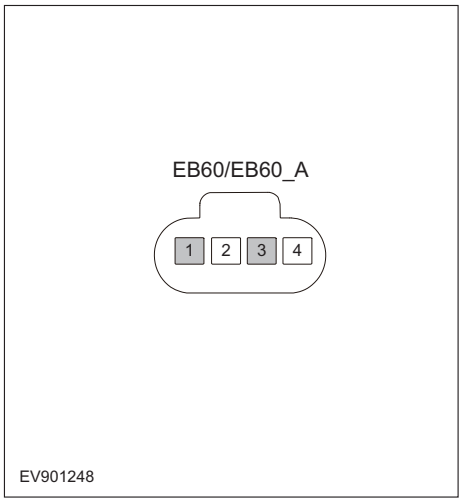

EB1000_2




EV901207

| Troubleshooting steps | | |
|---|---|---|
| Step 1 | General inspection. | |
| | | 1 Check the fuse EF12 for damage.<br>2 Check battery capacity.<br>3 Check the harness plug of charge indicator, VCU for damage, poor contact, aging and looseness.<br>4 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 2 | Check fault codes. | |
| | | 1 Connect the fault diagnosis instrument to the diagnosis interface.<br>2 Turn the ignition switch to ON position.<br>3 Diagnose the VCU with diagnostic apparatus.<br>4 Is there any other fault code except for P1CD7/P1CD8/P1CD9? |
| | | **YES →** Refer to:DTC Summary list(VCU). |
| **NO ↓** | | |
| Step 3 | Check the supply power of charge indicator when it is started or operating. | |
|  | | 1 Turn the ignition switch to OFF.<br>2 Disconnect the harness plug EB60/ EB60_A of charge indicator.<br>3 Turn the ignition switch to ON.<br>4 Measure the voltage between the harness terminal of charge indicator and ground.<br>Measuring circuit: voltage between terminal 1 on plug EB60/ EB60_A and ground.<br>**Standard value:10~14V**<br>5 Check whether the result is normal or not? |
| | | **NO →** Repair the circuit. |
| **YES ↓** | | |

| Step 4 | Check the circuit between charge indicator and VCU (open circuit). |
|---|---|
|  | 1 Turn the ignition switch to OFF.<br>2 Disconnect the harness plug EB60/ EB60_A of charge indicator.<br>3 Disconnect the harness plug EB1000_2 of GAW module.<br>4 Measure the resistance between the harness terminal of charge indicator and the VCU.<br>Measuring circuit: resistance between terminal 3 on plug EB60/ EB60_A and terminal B2 on plug EB1000_2.<br>**Standard value: ≈ 0Ω**<br>5 Check whether the result is normal or not? |
| | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | |

| Step 5 | Check the circuit of charge indicator (short circuit). |
|---|---|
|  | 1 Turn the ignition switch to OFF.<br>2 Measure the resistance between the harness terminal of charge indicator and ground.<br>Measuring circuit: resistance between terminal 1 on plug EB60/ EB60_A and ground.<br>Measuring circuit: resistance between terminal 3 on plug EB60/ EB60_A and ground.<br>**Standard value: ∞**<br>3 Measure the voltage between the harness terminal of charge indicator and ground.<br>Measuring circuit: voltage between terminal 3 on plug EB60/ EB60_A and ground.<br>**Standard value: ≈ 0V**<br>4 Check whether the result is normal or not? |
| | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | |

| Step 6 | Test whether VCU is norm. |
|---|---|
| | 1 Turn the ignition switch to OFF.<br>2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU.<br>3 Check and repair.<br>• Corrosion(install new adapting piece or terminal-cleaning module and pin).<br>• Pin damaged or bent-install new terminal/pin.<br>• Fly out of pins-install new pins as needed.<br>4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.<br>5 Run the system and determine if the problem persists? |
| | **YES →** Change the VCU Module |
| **NO ↓** | |
| At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems. | |
| **Next Step ↓** | |
| Diagnosis end. | |